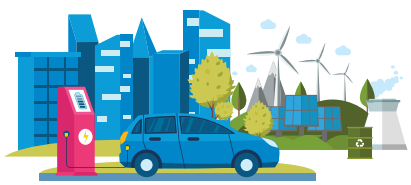
Building a Smarter, Sustainable Future: Next-Gen Research



> "Research is to see what everybody else has seen and to think what nobody else has thought."
>
> **Albert Szent-Györgyi**
> **Nobel Laureate, Physiology and Medicine, 1937**

Research and Development is at the heart of any society that aspires to create a smart and sustainable future. While India witnesses an unmistakable uptrend in startups and innovation, the fundamental drivers *(e.g., quest for knowledge, curiosity about the unknown and dare for imagination)* remain as important as ever. Hitachi Research in India is focused on those fundamentals. It is always looking for the new, challenging the old, and ready to disrupt and self-disrupt, for a better tomorrow.

Hitachi, an engineering major, has a rich legacy of more than nine decades of presence in India. Lately, it has embarked on a crucial journey of combining Operational and Information Technologies (OT x IT) for a wide spectrum of Hitachi Products. Artificial Intelligence (AI) and Machine Learning (ML) are some of the key enablers of that journey. Hitachi India R&D is a significant contributor to those AI/ML interventions. It brings about innovative solutions in the areas of Energy, Mobility and Digital. It has an unwavering focus on quality and always strives to strike a balance between Humans and Machines for a sustainable society.



Green

Electric Vehicle (EV) Charging

Charging payment solutions that also provide optimized recommendations for consumers, providers and ecosystem partners.

EV-Battery-Solutions-as-a-Service

Battery services for remaining useful life prediction, range estimation, charging recommendations, route planning, fleet operations and scheduling.

Vibration Analytics

For optimized production settings in wind and solar facilities, data-driven and real-time.

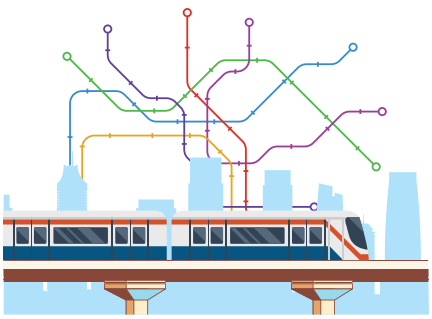
Green Building

Smart facility management using internet-of-things, mathematical modeling and simulation for smart administration, workflow management and energy optimization.

Carbon Accounting

Solutions for tracking, tracing and managing carbon footprints of an enterprise, across different phases of its operational lifecycle, value-chain and ecosystem partners.

KEY RESEARCH AREAS



Mobility

Railways Signalling and Safety

Solutions for the signalling systems of railway networks, metro rail included, with relentless focus on 100% operational safety.

Feeder Network Optimization

Optimization of the large transport networks holistically, along with feeder subsystems using mathematical optimization, simulation, computer vision, acoustics and sensors.

Mass Transit System

Solutions that deploy smart, friction-less payment systems for fast and smooth operations.



Digital

Artificial Intelligence and Generative AI

Solutions that cover multiple domains like energy, mobility, cyber-security, smart spaces, precision agriculture and industrial processes.

Financial Services

Solutions that align with FinTech expansion in India, and specifically for merchant ecosystem of the banking partners of Hitachi's Payment Services.

Cyber-securtity

Solutions for edge-device security and fuzzing platforms to detect zero-day attack vulnerabilities.

Human Computer Interaction

Next-generation human computer interfaces for industrial and social innovation systems.

Voice and Speech Processing

For digital and financial inclusion that leverage voice authentication and automated speech recognition for Indian vernacular languages.

Computer Vision

Solutions for industrial quality control, Advanced Driving Assistance (ADAS), social innovation, event detection and safety surveillance.